

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250273118

Kind Code

A1

Publication Date

August 28, 2025

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DISPLAY APPARATUS, METHOD OF DRIVING DISPLAY PANEL USING THE SAME AND ELECTRONIC APPARATUS INCLUDING THE SAME

Abstract

A display apparatus includes a display panel, a data driver and a driving controller. The display panel includes a plurality of display blocks. The data driver is configured to output data voltages to the display panel. The driving controller is configured to control the data driver. The driving controller is configured to calculate differences between loads of a previous frame and loads of a present frame for the display blocks for each colors, to determine whether an image is a static image or not based on the differences between the loads of the previous frame and the loads of the present frame for the display blocks for the each colors, and to reduce luminance of the image when the image is determined as the static image.

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Family ID: 1000008324655

Appl. No.: 18/966125

Filed: December 03, 2024

Foreign Application Priority Data

KR 10-2024-0029274

Feb. 28, 2024

Publication Classification

Int. Cl.: G09G3/20 (20060101)

U.S. Cl.:

CPC **G09G3/2092** (20130101); G09G2310/0275 (20130101); G09G2320/0233 (20130101); G09G2320/0257 (20130101); G09G2320/043 (20130101); G09G2320/0626 (20130101)

Background/Summary

PRIORITY STATEMENT

[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0029274, filed on Feb. 28, 2024, in the Korean Intellectual Property Office KIPO, the contents of which are herein incorporated by reference in their entireties.

BACKGROUND

1. Field

[0002] Embodiments of the present inventive concept relate to a display apparatus, a method of driving a display panel using the display apparatus and an electronic apparatus including the display apparatus. More particularly, embodiments of the present inventive concept relate to a display apparatus determining a static image based on differences between loads of a previous frame and loads of a present frame for display blocks for each colors of the display panel to enhance an accuracy of a determination of the static image, a method of driving a display panel using the display apparatus and an electronic apparatus including the display apparatus.

2. Description of the Related Art

[0003] Generally, a display apparatus includes a display panel and a display panel driver. The display panel displays an image based on input image data. The display panel includes a plurality of gate lines, a plurality of data lines and a plurality of pixels. The display panel driver includes a gate driver, a data driver and a driving controller. The gate driver outputs gate signals to the gate lines. The data driver outputs data voltages to the data lines. The driving controller controls an operation of the gate driver and an operation of the data driver.

[0004] When the same image is displayed on the display panel for a long time, an afterimage may remain on the display panel. A shape of the afterimage may be determined according to a content of the input image data. Due to the afterimage, a display quality of the display panel may be deteriorated.

[0005] To prevent the deterioration of the display quality due to the afterimage, the driving controller may decrease a luminance of a display image when the same image is displayed on the display panel for a long time. When the luminance of the display image is decreased, the afterimage of the display panel may be prevented and the power consumption of the display apparatus may be reduced. However, a moving image with little movements between frames and a moving image which is dark overall may be mistakenly determined as a static image so that an inconvenience may be occurred to a user.

SUMMARY

[0006] Embodiments of the present inventive concept provide a display apparatus determining whether an image is a static image or not based on differences between loads of a previous frame and loads of a present frame for display blocks for each colors of the display panel to enhance an accuracy of a determination of the static image.

[0007] Embodiments of the present inventive concept also provide a method of driving a display panel using the display apparatus.

[0008] Embodiments of the present inventive concept also provide an electronic apparatus including the display apparatus.

[0009] In an embodiment of a display apparatus according to the present inventive concept, the

display apparatus includes a display panel, a data driver and a driving controller. The display panel includes a plurality of display blocks. The data driver is configured to output data voltages to the display panel. The driving controller is configured to control the data driver. The driving controller is configured to calculate differences between loads of a previous frame and loads of a present frame for the plurality of display blocks for each colors, to determine whether an image is a static image or not based on the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors, and to reduce luminance of the image when the image is determined as the static image.

[0010] In an embodiment, the driving controller may be configured to determine a maximum value among absolute values of the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors.

[0011] In an embodiment, when a load of an X-1-th frame of a first color in a first display block is $LOAD(X-1, R1)$, a load of an X-th frame of the first color in the first display block is $LOAD(X, R1)$, a load of the X-1-th frame of the first color in a second display block is $LOAD(X-1, R2)$, a load of the X-th frame of the first color in the second display block is $LOAD(X, R2)$, a load of the X-1-th frame of the first color in an N-th display block is $LOAD(X-1, RN)$, a load of the X-th frame of the first color in the N-th display block is $LOAD(X, RN)$, a load of the X-1-th frame of a second color in the first display block is $LOAD(X-1, G1)$, a load of the X-th frame of the second color in the first display block is $LOAD(X, G1)$, a load of the X-1-th frame of the second color in the second display block is $LOAD(X-1, G2)$, a load of the X-th frame of the second color in the second display block is $LOAD(X, G2)$, a load of the X-1-th frame of the second color in the N-th display block is $LOAD(X-1, GN)$, a load of the X-th frame of the second color in the N-th display block is $LOAD(X, GN)$, a load of the X-1-th frame of a third color in the first display block is $LOAD(X-1, B1)$, a load of the X-th frame of the third color in the first display block is $LOAD(X, B1)$, a load of the X-1-th frame of the third color in the second display block is $LOAD(X-1, B2)$, a load of the X-th frame of the third color in the second display block is $LOAD(X, B2)$, a load of the X-1-th frame of the third color in the N-th display block is $LOAD(X-1, BN)$, a load of the X-th frame of the third color in the N-th display block is $LOAD(X, BN)$, the maximum value may be $\text{MAX}(|LOAD(X-1, R1)-LOAD(X, R1)|, |LOAD(X-1, R2)-LOAD(X, R2)|, \dots, |LOAD(X-1, RN)-LOAD(X, RN)|, |LOAD(X-1, G1)-LOAD(X, G1)|, |LOAD(X-1, G2)-LOAD(X, G2)|, \dots, |LOAD(X-1, GN)-LOAD(X, GN)|, |LOAD(X-1, B1)-LOAD(X, B1)|, |LOAD(X-1, B2)-LOAD(X, B2)|, \dots, |LOAD(X-1, BN)-LOAD(X, BN)|)$.

[0012] In an embodiment, the driving controller may be configured to generate a sensitivity value by normalizing the maximum value to a ratio of a difference between a maximum load and a minimum load.

[0013] In an embodiment, when the sensitivity value is Δ , an absolute value of a difference between a load of an X-1-th frame of a first color in a first display block and a load of an X-th frame of the first color in the first display block is $D(R1)$, an absolute value of a difference between a load of the X-1-th frame of the first color in a second display block and a load of the X-th frame of the first color in the second display block is $D(R2)$, an absolute value of a difference between a load of the X-1-th frame of the first color in an N-th display block and a load of the X-th frame of the first color in the N-th display block is $D(RN)$, an absolute value of a difference between a load of the X-1-th frame of a second color in the first display block and a load of the X-th frame of the second color in the first display block is $D(G1)$, an absolute value of a difference between a load of the X-1-th frame of the second color in the second display block and a load of the X-th frame of the second color in the second display block is $D(G2)$, an absolute value of a difference between a load of the X-1-th frame of the second color in the N-th display block and a load of the X-th frame of the second color in the N-th display block is $D(GN)$, an absolute value of a difference between a load of the X-1-th frame of a third color in the first display block and a load of the X-th frame of the third color in the first display block is $D(B1)$, an absolute value of a difference between a load

of the X-1-th frame of the third color in the second display block and a load of the X-th frame of the third color in the second display block is D(B2), an absolute value of a difference between a load of the X-1-th frame of the third color in the N-th display block and a load of the X-th frame of the third color in the N-th display block is D(BN), the maximum load is MAXLO, the minimum load is MINLO and a number of the display blocks of the display panel is N,

$$[00001] = \frac{\text{MAX}(D(R1), D(R2), \dots, D(RN), D(G1), D(G2), \dots, D(GN), D(B1), D(B2), \dots, D(BN))}{(\text{MAXLO} - \text{MINLO}) / (N \times 3)}$$

may be satisfied.

[0014] In an embodiment, when the sensitivity value is less than a first threshold value, the driving controller may be configured to determine data of the present frame as static image data.

[0015] In an embodiment, when a maintaining time of the static image is greater than a second threshold value, the driving controller may be configured to reduce the luminance of the image

[0016] In an embodiment, the driving controller may include a load calculator configured to calculate the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors, a load storer configured to store the loads of the previous frame for the plurality of display blocks for the each colors and a static image determiner configured to receive the loads of the present frame for the plurality of display blocks for the each colors from the load calculator and the loads of the previous frame for the plurality of display blocks for the each colors from the load storer and to determine whether the image is the static image or not based on the differences between the loads of the previous frame for the plurality of display blocks for the each colors and the loads of the present frame for the plurality of display blocks for the each colors.

[0017] In an embodiment, the driving controller may further include an image compensator configured to receive a flag signal indicating the image as the static image from the static image determiner and to reduce the luminance when the image is determined as the static image.

[0018] In an embodiment of a method of driving a display panel according to the present inventive concept, the method includes calculating differences between loads of a previous frame and loads of a present frame for display blocks for each colors in the display panel, determining whether an image is a static image or not based on the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors, reducing luminance of the image when the image is determined as the static image to generate compensated image data and generating data voltages based on the compensated image data and outputting the data voltages to the display panel.

[0019] In an embodiment, the determining whether the image is a static image or not may include determining a maximum value among absolute values of the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors.

[0020] In an embodiment, the determining whether the image is a static image or not may further include generating a sensitivity value by normalizing the maximum value to a ratio of a difference between a maximum load and a minimum load.

[0021] In an embodiment, the determining whether the image is a static image or not may further include determining data of the present frame as static image data when the sensitivity value is less than a first threshold value.

[0022] In an embodiment, the reducing luminance of the image when the imaged is determined as the static image may include reducing the luminance of the image when a maintaining time of the static image is greater than a second threshold value.

[0023] In an embodiment of a display apparatus according to the present inventive concept, the display apparatus includes a display panel, a data driver and a driving controller. The display panel includes a plurality of display blocks. The data driver is configured to output data voltages to the display panel. The driving controller is configured to control the data driver. The driving controller

is configured to calculate differences between currents or luminances of a previous frame and currents or luminances of a present frame for the plurality of display blocks for each colors, to determine whether an image is a static image or not based on the differences between the currents or luminances of the previous frame and the currents or luminances of the present frame for the plurality of display blocks for the each colors, and to reduce luminance of the image when the image is determined as the static image.

[0024] In an embodiment, the driving controller may be configured to determine a maximum value among absolute values of the differences between the currents or luminances of the previous frame and the currents or luminances of the present frame for the plurality of display blocks for the each colors.

[0025] In an embodiment, the driving controller may be configured to generate a sensitivity value by normalizing the maximum value to a ratio of a difference between a maximum current or luminance and a minimum current or luminance.

[0026] In an embodiment, when the sensitivity value is less than a first threshold value, the driving controller may be configured to determine data of the present frame as static image data.

[0027] In an embodiment, the driving controller may include a current calculator configured to calculate the currents of the previous frame and the currents of the present frame for the plurality of display blocks for the each colors, a current storer configured to store the currents of the previous frame for the plurality of display blocks for the each colors and a static image determiner configured to receive the currents of the present frame for the plurality of display blocks for the each colors from the current calculator and the currents of the previous frame for the plurality of display blocks for the each colors from the current storer and to determine the image as the static image based on the differences between the currents of the previous frame for the plurality of display blocks for the each colors and the currents of the present frame for the plurality of display blocks for the each colors.

[0028] In an embodiment, the driving controller may include a luminance calculator configured to calculate the luminances of the previous frame and the luminances of the present frame for the plurality of display blocks for the each colors, a luminance storer configured to store the luminances of the previous frame for the plurality of display blocks for the each colors and a static image determiner configured to receive the luminances of the present frame for the plurality of display blocks for the each colors from the luminance calculator and the luminances of the previous frame for the plurality of display blocks for the each colors from the luminance storer and to determine the image as the static image based on the differences between the luminances of the previous frame for the plurality of display blocks for the each colors and the luminances of the present frame for the plurality of display blocks for the each colors.

[0029] In an embodiment of an electronic apparatus according to the present inventive concept, the electronic apparatus includes a display panel, a data driver, a driving controller and a processor. The display panel includes a plurality of display blocks. The data driver is configured to output data voltages to the display panel. The driving controller is configured to control the data driver. The processor is configured to output input image data and an input control signal to the driving controller. The driving controller is configured to calculate differences between loads of a previous frame and loads of a present frame for the plurality of display blocks for each colors, to determine whether an image is a static image or not based on the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors, and to reduce luminance of the image when the image is determined as the static image.

[0030] According to the display apparatus, the method of driving the display panel using the display apparatus and the electronic apparatus including the display apparatus, the image is determined as the static image based on the differences between the loads of the previous frame and the loads of the present frame for the display blocks for the each colors of the display panel to enhance an accuracy of the determination of the static image.

[0031] Alternatively, the image is determined as the static image based on the differences between the currents of the previous frame and the currents of the present frame for the display blocks for the each colors of the display panel to enhance an accuracy of the determination of the static image.

[0032] Alternatively, the image is determined as the static image based on the differences between the luminances of the previous frame and the luminances of the present frame for the display blocks for the each colors of the display panel to enhance an accuracy of the determination of the static image.

[0033] Accordingly, an error mistakenly determining a moving image with little movements between frames and a moving image which is dark overall as a static image may be prevented.

[0034] Therefore, when the same image is displayed on the display panel for a long time, the luminance of the display image may be reduced so that the afterimage of the display panel may be prevented and the power consumption of the display apparatus may be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0035] The above and other features and advantages of the present inventive concept will become more apparent by describing in detailed embodiments thereof with reference to the accompanying drawings, in which:

[0036] FIG. 1 is a block diagram illustrating a display apparatus according to an embodiment of the present inventive concept;

[0037] FIG. 2 is a diagram illustrating an example of display blocks of a display panel of FIG. 1;

[0038] FIG. 3 is a diagram illustrating first color image data of an X-1-th frame for the display blocks of the display panel of FIG. 1;

[0039] FIG. 4 is a diagram illustrating second color image data of the X-1-th frame for the display blocks of the display panel of FIG. 1;

[0040] FIG. 5 is a diagram illustrating third color image data of the X-1-th frame for the display blocks of the display panel of FIG. 1;

[0041] FIG. 6 is a diagram illustrating first color image data of an X-th frame for the display blocks of the display panel of FIG. 1;

[0042] FIG. 7 is a diagram illustrating second color image data of the X-th frame for the display blocks of the display panel of FIG. 1;

[0043] FIG. 8 is a diagram illustrating third color image data of the X-th frame for the display blocks of the display panel of FIG. 1;

[0044] FIG. 9 is a block diagram illustrating a driving controller of FIG. 1;

[0045] FIG. 10 is a flowchart diagram illustrating an operation of the driving controller of FIG. 1;

[0046] FIG. 11 is a graph illustrating an example of a determination of a static image according to a comparative embodiment;

[0047] FIG. 12 is a graph illustrating an example of a determination of a static image according to the present embodiment;

[0048] FIG. 13 is a graph illustrating an example of a determination of a static image according to a comparative embodiment;

[0049] FIG. 14 is a graph illustrating an example of a determination of a static image according to the present embodiment;

[0050] FIG. 15 is a block diagram illustrating a driving controller of a display apparatus according to an embodiment of the present inventive concept;

[0051] FIG. 16 is a block diagram illustrating a driving controller of a display apparatus according to an embodiment of the present inventive concept;

[0052] FIG. 17 is a block diagram illustrating an electronic apparatus according to an embodiment

of the present inventive concept;

[0053] FIG. **18** is a diagram illustrating an example in which the electronic apparatus of FIG. **17** is implemented as a smartphone;

[0054] FIG. **19** is a diagram illustrating an example in which the electronic apparatus of FIG. **17** is implemented as a monitor; and

[0055] FIG. **20** is a block diagram illustrating an electronic apparatus according to an embodiment of the present inventive concept.

DETAILED DESCRIPTION OF THE INVENTIVE CONCEPT

[0056] Hereinafter, the present inventive concept will be explained in detail with reference to the accompanying drawings.

[0057] FIG. **1** is a block diagram illustrating a display apparatus according to an embodiment of the present inventive concept.

[0058] Referring to FIG. **1**, the display apparatus includes a display panel **100** and a display panel driver. The display panel driver drives the display panel **100**. The display panel driver includes a driving controller **200**, a gate driver **300**, a gamma reference voltage generator **400** and a data driver **500**.

[0059] For example, the driving controller **200** and the data driver **500** may be integrally formed. The driving controller **200**, the gamma reference voltage generator **400** and the data driver **500** may be integrally formed. A driving module in which at least the driving controller **200** and the data driver **500** are integrated may be a timing controller embedded data driver (TED).

[0060] The display panel **100** has a display region AA on which an image is displayed and a peripheral region PA disposed adjacent to the display region AA.

[0061] The display panel **100** includes a plurality of gate lines GL, a plurality of data lines DL and a plurality of pixels P connected to the gate lines GL and the data lines DL. The gate lines GL may extend in a first direction D1 and the data lines DL may extend in a second direction D2 crossing the first direction D1.

[0062] The driving controller **200** receives input image data IMG and an input control signal CONT from an external apparatus (e.g., an application processor). For example, the input image data IMG may include red image data, green image data and blue image data. For example, the input image data IMG may include white image data. For example, the input image data IMG may include magenta image data, yellow image data and cyan image data. The input control signal CONT may include a master clock signal and a data enable signal. The input control signal CONT may further include a vertical synchronizing signal and a horizontal synchronizing signal.

[0063] The driving controller **200** generates a first control signal CONT1, a second control signal CONT2, a third control signal CONT3 and a data signal DATA based on the input image data IMG and the input control signal CONT.

[0064] The driving controller **200** generates the first control signal CONT1 for controlling an operation of the gate driver **300** based on the input control signal CONT, and outputs the first control signal CONT1 to the gate driver **300**. The first control signal CONT1 may include a vertical start signal and a gate clock signal.

[0065] The driving controller **200** generates the second control signal CONT2 for controlling an operation of the data driver **500** based on the input control signal CONT, and outputs the second control signal CONT2 to the data driver **500**. The second control signal CONT2 may include a horizontal start signal and a load signal.

[0066] The driving controller **200** generates the data signal DATA based on the input image data IMG. The driving controller **200** outputs the data signal DATA to the data driver **500**.

[0067] The driving controller **200** generates the third control signal CONT3 for controlling an operation of the gamma reference voltage generator **400** based on the input control signal CONT, and outputs the third control signal CONT3 to the gamma reference voltage generator **400**.

[0068] The gate driver **300** generates gate signals driving the gate lines GL in response to the first

control signal CONT1 received from the driving controller **200**. The gate driver **300** outputs the gate signals to the gate lines GL. For example, the gate driver **300** may sequentially output the gate signals to the gate lines GL. For example, the gate driver **300** may be mounted on the peripheral region PA of the display panel **100**. For example, the gate driver **300** may be integrated on the peripheral region PA of the display panel **100**.

[0069] The gamma reference voltage generator **400** generates a gamma reference voltage VGREF in response to the third control signal CONT3 received from the driving controller **200**. The gamma reference voltage generator **400** provides the gamma reference voltage VGREF to the data driver **500**.

[0070] In an embodiment, the gamma reference voltage generator **400** may be disposed in the driving controller **200**, or in the data driver **500**.

[0071] The data driver **500** receives the second control signal CONT2 and the data signal DATA from the driving controller **200**, and receives the gamma reference voltages VGREF from the gamma reference voltage generator **400**. The data driver **500** converts the data signal DATA into data voltages having an analog type using the gamma reference voltages VGREF. The data driver **500** outputs the data voltages to the data lines DL.

[0072] FIG. 2 is a diagram illustrating an example of display blocks of the display panel **100** of FIG. 1.

[0073] Referring to FIGS. 1 and 2, the display panel **100** may include a plurality of display blocks.

[0074] For example, each of the display blocks may extend in the second direction D2. Each of the display blocks may be a rectangular shape having a shorter side extending in the first direction D1 and a longer side extending in the second direction D2. The display blocks may be disposed in the first direction D1.

[0075] In FIG. 2, the display panel **100** may include a first display block A1, a second display block A2, a third display block A3, . . . , an N-2-th display block AN-2, an N-1-th display block AN-1 and an N-th display block AN. For example, a number of the display blocks may be equal to or greater than ten. For example, the number of the display blocks may be sixteen.

[0076] FIG. 3 is a diagram illustrating first color image data of an X-1-th frame FRAME X-1 for the display blocks of the display panel **100** of FIG. 1. FIG. 4 is a diagram illustrating second color image data of the X-1-th frame FRAME X-1 for the display blocks of the display panel **100** of FIG. 1. FIG. 5 is a diagram illustrating third color image data of the X-1-th frame FRAME X-1 for the display blocks of the display panel **100** of FIG. 1. FIG. 6 is a diagram illustrating first color image data of an X-th frame FRAME X for the display blocks of the display panel **100** of FIG. 1. FIG. 7 is a diagram illustrating second color image data of the X-th frame FRAME X for the display blocks of the display panel **100** of FIG. 1. FIG. 8 is a diagram illustrating third color image data of the X-th frame FRAME X for the display blocks of the display panel **100** of FIG. 1.

[0077] Referring to FIGS. 1 to 8, for example, the first color image data may be red image data. For example, the second color image data may be green image data. For example, the third color image data may be blue image data.

[0078] As shown in FIG. 3, the first color image data of the X-1-th frame FRAME X-1 may include first block red data R1, second block red data R2, third block red data R3, . . . , N-2-th block red data RN-2, N-1-th block red data RN-1 and N-th block red data RN.

[0079] As shown in FIG. 4, the second color image data of the X-1-th frame FRAME X-1 may include first block green data G1, second block green data G2, third block green data G3, . . . , N-2-th block green data GN-2, N-1-th block green data GN-1 and N-th block green data GN.

[0080] As shown in FIG. 5, the third color image data of the X-1-th frame FRAME X-1 may include first block blue data B1, second block blue data B2, third block blue data B3, . . . N-2-th block blue data BN-2, N-1-th block blue data BN-1 and N-th block blue data BN.

[0081] As shown in FIG. 6, the first color image data of the X-th frame FRAME X may include first block red data R1, second block red data R2, third block red data R3, . . . , N-2-th block red

data RN-2, N-1-th block red data RN-1 and N-th block red data RN.

[0082] As shown in FIG. 7, the second color image data of the X-th frame FRAME X may include first block green data G1, second block green data G2, third block green data G3, . . . , N-2-th block green data GN-2, N-1-th block green data GN-1 and N-th block green data GN.

[0083] As shown in FIG. 8, the third color image data of the X-th frame FRAME X may include first block blue data B1, second block blue data B2, third block blue data B3, . . . N-2-th block blue data BN-2, N-1-th block blue data BN-1 and N-th block blue data BN.

[0084] FIG. 9 is a block diagram illustrating the driving controller 200 of FIG. 1. FIG. 10 is a flowchart diagram illustrating an operation of the driving controller 200 of FIG. 1.

[0085] Referring to FIGS. 1 to 10, the display apparatus includes the display panel 100, the data driver 500 and the driving controller 200.

[0086] The driving controller 200 controls the data driver 500. The driving controller 200 may calculate differences between loads LOAD(X-1) of the previous frame and loads LOAD(X) of the present frame for the display blocks for each colors. The driving controller 200 may determine a static image based on the differences between the loads LOAD(X-1) of the previous frame and the loads LOAD(X) of the present frame for the display blocks for the each colors. The driving controller 200 may reduce a luminance for the static image.

[0087] For example, the driving controller 200 may determine a maximum value among absolute values of the differences between the loads LOAD(X-1) of the previous frame and the loads LOAD(X) of the present frame for the display blocks for the each colors.

[0088] Herein, when the load of the X-1-th frame of the first color in the first display block is LOAD(X-1, R1), the load of the X-th frame of the first color in the first display block is LOAD(X, R1), the load of the X-1-th frame of the first color in the second display block is LOAD(X-1, R2), the load of the X-th frame of the first color in the second display block is LOAD(X, R2), the load of the X-1-th frame of the first color in the N-th display block is LOAD(X-1, RN), the load of the X-th frame of the first color in the N-th display block is LOAD(X, RN), the load of the X-1-th frame of the second color in the first display block is LOAD(X-1, G1), the load of the X-th frame of the second color in the first display block is LOAD(X, G1), the load of the X-1-th frame of the second color in the second display block is LOAD(X-1, G2), the load of the X-th frame of the second color in the second display block is LOAD(X, G2), the load of the X-1-th frame of the second color in the N-th display block is LOAD(X-1, GN), the load of the X-th frame of the second color in the N-th display block is LOAD(X, GN), the load of the X-1-th frame of the third color in the first display block is LOAD(X-1, B1), the load of the X-th frame of the third color in the first display block is LOAD(X, B1), the load of the X-1-th frame of the third color in the second display block is LOAD(X-1, B2), the load of the X-th frame of the third color in the second display block is LOAD(X, B2), the load of the X-1-th frame of the third color in the N-th display block is LOAD(X-1, BN), the load of the X-th frame of the third color in the N-th display block is LOAD(X, BN), the maximum value may be MAX(|LOAD(X-1, R1)-LOAD(X, R1)|, |LOAD(X-1, R2)-LOAD(X, R2)|, . . . , |LOAD(X-1, RN)-LOAD(X, RN)|, |LOAD(X-1, G1)-LOAD(X, G1)|, |LOAD(X-1, G2)-LOAD(X, G2)|, . . . , |LOAD(X-1, GN)-LOAD(X, GN)|, |LOAD(X-1, B1)-LOAD(X, B1)|, |LOAD(X-1, B2)-LOAD(X, B2)|, . . . , |LOAD(X-1, BN)-LOAD(X, BN)|).

[0089] The driving controller 200 may generate a sensitivity value by normalizing the maximum value to a ratio of a difference between a maximum load and a minimum load. The maximum value is a maximum absolute value among the absolute values of the differences between the loads LOAD(X-1) of the previous frame and the loads LOAD(X) of the present frame. When the maximum value is normalized by the ratio of the difference between the maximum load and the minimum load, the difference between the load LOAD(X-1) of the previous frame and the load LOAD(X) of the present frame is not an absolute value but a relative value with respect to the difference between the maximum load and the minimum load.

[0090] When the sensitivity value is Δ , an absolute value of the difference between the load of the X-1-th frame of the first color in the first display block and the load of the X-th frame of the first color in the first display block is D(R1), an absolute value of the difference between the load of the X-1-th frame of the first color in the second display block and the load of the X-th frame of the first color in the second display block is D(R2), an absolute value of the difference between the load of the X-1-th frame of the first color in the N-th display block and the load of the X-th frame of the first color in the N-th display block is D(RN), an absolute value of the difference between the load of the X-1-th frame of the second color in the first display block and the load of the X-th frame of the second color in the first display block is D(G1), an absolute value of the difference between the load of the X-1-th frame of the second color in the second display block and the load of the X-th frame of the second color in the second display block is D(G2), an absolute value of the difference between the load of the X-1-th frame of the second color in the N-th display block and the load of the X-th frame of the second color in the N-th display block is D(GN), an absolute value of the difference between the load of the X-1-th frame of the third color in the first display block and the load of the X-th frame of the third color in the first display block is D(B1), an absolute value of the difference between the load of the X-1-th frame of the third color in the second display block and the load of the X-th frame of the third color in the second display block is D(B2), an absolute value of the difference between the load of the X-1-th frame of the third color in the N-th display block and the load of the X-th frame of the third color in the N-th display block is D(BN), the maximum load is MAXLO, the minimum load is MINLO and the number of the display blocks of the display panel is N,

$$[00002] = \frac{\text{MAX}(D(R1), D(R2), \dots, D(RN), D(G1), D(G2), \dots, D(GN), D(B1), D(B2), \dots, D(BN))}{(\text{MAXLO} - \text{MINLO}) / (N \times 3)}$$

may be satisfied.

[0091] Herein, $D(R1)=|\text{LOAD}(X-1, R1)-\text{LOAD}(X, R1)|$, $D(R2)=|\text{LOAD}(X-1, R2)-\text{LOAD}(X, R2)|$, $D(RN)=|\text{LOAD}(X-1, RN)-\text{LOAD}(X, RN)|$, $D(G1)=|\text{LOAD}(X-1, G1)-\text{LOAD}(X, G1)|$, $D(G2)=|\text{LOAD}(X-1, G2)-\text{LOAD}(X, G2)|$, $D(GN)=|\text{LOAD}(X-1, GN)-\text{LOAD}(X, GN)|$, $D(B1)=|\text{LOAD}(X-1, B1)-\text{LOAD}(X, B1)|$, $D(B2)=|\text{LOAD}(X-1, B2)-\text{LOAD}(X, B2)|$, $D(BN)=|\text{LOAD}(X-1, BN)-\text{LOAD}(X, BN)|$.

[0092] Herein, the maximum load MAXLO and the minimum load MINLO may be values for the entire display panel **100**, not for each display block. In addition, the maximum load MAXLO and the minimum load MINLO may be values for all colors, not for each color.

[0093] The maximum value $\text{MAX}(D(R1), D(R1), \dots, D(RN), D(G1), D(G2), \dots, D(GN), D(B1), D(B2), \dots, D(BN))$ disposed in a numerator when calculating the sensitivity value Δ is the load difference corresponding to one display block and one color so that the difference between the maximum load MAXLO and the minimum load MINLO in a denominator may be divided by $(N \times 3)$, where “N” is the number of the display blocks of the display panel **100** and “3” is the number of colors of the input image data IMG.

[0094] When the sensitivity value Δ is less than a first threshold value TH, the driving controller **200** may determine data IMG(X) of the present frame as static image data.

[0095] When a maintaining time of the static image is greater than a second threshold value TTH, the driving controller **200** may reduce the luminance for the static image.

[0096] Referring to FIG. 9, the driving controller **200** may include a load calculator **220** calculating the loads $\text{LOAD}(X-1)$ of the previous frame and the loads $\text{LOAD}(X)$ of the present frame for the display blocks for the each colors, a load storer **240** storing the loads $\text{LOAD}(X-1)$ of the previous frame for the display blocks for the each colors and a static image determiner **260** receiving the loads $\text{LOAD}(X)$ of the present frame for the display blocks for the each colors from the load calculator **220** and the loads $\text{LOAD}(X-1)$ of the previous frame for the display blocks for the each colors from the load storer **240** and determining whether the image is a static image or not based on

the differences between the loads $LOAD(X-1)$ of the previous frame for the display blocks for the each colors and the loads $LOAD(X)$ of the present frame for the display blocks for the each colors. [0097] The driving controller **200** may further include an image compensator **280** receiving a flag signal FL indicating the static image from the static image determiner **260** and reducing the luminance for the input image data IMG(X).

[0098] Referring to FIG. **10**, in a method of driving the display panel **100** according to the present embodiment, the driving controller **200** may convert the grayscale value of the input image data IMG to the load (operation S**100**). For example, the driving controller **200** may convert the grayscale value of the input image data IMG to the load by a gamma conversion. The load of the input image data IMG corresponds to a luminance actually perceived by a user's eyes. Accordingly, when the image displayed on the display panel is determined as the static image using the load of the input image data IMG, the accuracy of the determination of the static image may be enhanced compared to a case in which the static image is determined using the grayscale value of the input image data IMG.

[0099] The driving controller **200** may divide the load of the input image data IMG into N blocks (operation S**200**).

[0100] The driving controller **200** may calculate the difference between the loads $LOAD(X-1)$ of the previous frame and the loads $LOAD(X)$ of the present frame for the N blocks for each colors (operation S**300**).

[0101] When calculating one load value for each display block for one color in each frame, the load values of all pixels of the one color in each display block may be added up. When calculating one load value for each display block for the one color in each frame, one frame data may only occupy a memory space equal to a multiplication of the number of the display blocks by the number of colors. For example, when the number of the display blocks is N and the number of colors is three, the loads $LOAD(X-1)$ of the previous frame may have 3N values and the loads $LOAD(X)$ of the present frame may have 3N values. The load storer **240** may require only space to store 3N values.

[0102] Humans have two eyes in a horizontal direction so that humans are more sensitive to image movement in the horizontal direction than in a vertical direction. Thus, the display block has a rectangular shape extending in the vertical direction (e.g., the second direction D2).

[0103] The driving controller **200** may determine the maximum value among the absolute values of the differences between the loads $LOAD(X-1)$ of the previous frame and the loads $LOAD(X)$ of the present frame for the display blocks for the each colors (operation S**400**).

[0104] The driving controller **200** may calculate the sensitivity value Δ by normalizing the maximum value to a ratio of the difference between the maximum load and the minimum load (operation S**500**).

[0105] When the sensitivity value is less than the first threshold value TH, the driving controller **200** may determine the data IMG(X) of the present frame as the static image data (operation S**600**).

[0106] When the maintaining time of the static image is greater than the second threshold value TTH, the driving controller **200** may reduce the luminance for the static image (operation S**700**). If the luminance is reduced as soon as the image is determined as the static image, the luminance decrease may be applied instantly and also be released instantly so that the luminance decrease may be easily recognized to a user's eyes. When the luminance decrease is easily recognized to the user's eyes, the user may feel uncomfortable when using the display apparatus. Thus, the driving controller **200** may reduce the luminance when the maintaining time of the static image is greater than the second threshold value TTH. In addition, the luminance may be gradually reduced over a long period of time to ensure that the luminance decreases is not easily recognized to the user's eyes. The driving controller **200** may reduce the luminance for the static image to generate compensated image data CIMG(X). The data driver **500** may generate the data voltage based on the compensated image data CIMG(X) and output the data voltage to the display panel **100**.

[0107] FIG. **11** is a graph illustrating an example of a determination of a static image according to a

comparative embodiment. FIG. 12 is a graph illustrating an example of a determination of a static image according to the present embodiment.

[0108] In FIGS. 11 and 12, the determination of the static image according to the comparative embodiment and the determination of the static image according to the present embodiment are compared for the same image. FIGS. 11 and 12 represent a case in which a moving image with little movements between frames is displayed on the display panel.

[0109] In the comparative embodiment, the static image may be determined based on the difference between the load of the previous frame of the entire display panel and all colors, and the load of the present frame of the entire display panel and the all colors. In the present embodiment, the static image may be determined based on the differences between the loads of the previous frame and the loads of the present frame for the display blocks for the each colors.

[0110] In the comparative embodiment, the moving image with little movements between frames may be mistakenly determined by the static image. When the moving image is displayed on the display panel as disclosed in FIG. 11, only once, for example, after 2 minutes of a 3-minute playback time, the image displayed on the display panel is determined as the moving image.

[0111] On the other hand, in the present embodiment, the static image is determined using the absolute value of the difference between the loads of the previous frame and the loads of the present frame for the display blocks for the each colors so that the image displayed on the display panel is determined as a moving image once after approximately 20 seconds and once again approximately 30 seconds later. moving image is determined at approximately 20 to 30 seconds intervals.

[0112] FIG. 13 is a graph illustrating an example of a determination of a static image according to a comparative embodiment. FIG. 14 is a graph illustrating an example of a determination of a static image according to the present embodiment.

[0113] In FIGS. 13 and 14, the determination of the static image according to the comparative embodiment and the determination of the static image according to the present embodiment are compared for the same image. FIGS. 13 and 14 represent a case in which a moving image which is dark overall is displayed on the display panel.

[0114] In the comparative embodiment, the moving image which is dark overall may be mistakenly determined as the static image. Although only the moving image is displayed on the display panel in FIG. 13, a moving image is determined once after approximately 4 minutes and 30 seconds, once again approximately 3 minutes and 50 seconds later and once again approximately 2 minutes and 15 seconds in a 12-minute playback time.

[0115] On the other hand, in the present embodiment, the static image is determined using the absolute value of the difference between the loads of the previous frame and the loads of the present frame for the display blocks for the each colors so that a moving image may be determined in much shorter intervals.

[0116] According to the present embodiment, the static image may be determined based on the differences between the loads $LOAD(X-1)$ of the previous frame and the loads $LOAD(X)$ of the present frame for the display blocks for the each colors to enhance an accuracy of the determination of the static image.

[0117] Accordingly, an error mistakenly determining a moving image with little movements between frames and a moving image which is dark overall as a static image may be prevented.

[0118] Therefore, when the same image is displayed on the display panel for a long time, the luminance of the display image may decrease so that the afterimage of the display panel 100 may be prevented and the power consumption of the display apparatus may be reduced.

[0119] FIG. 15 is a block diagram illustrating a driving controller 200 of a display apparatus according to an embodiment of the present inventive concept.

[0120] The display apparatus according to the present embodiment is substantially the same as the display apparatus of the previous embodiment explained referring to FIGS. 1 to 14 except that the

driving controller determines the static image not based on the loads of the frames but based on currents of the frames. Thus, the same reference numerals will be used to refer to the same or like parts as those described in the previous embodiment of FIGS. **1** to **14** and any repetitive explanation concerning the above elements will be omitted.

[0121] Referring to FIGS. **1** to **8** and **10** to **15**, the driving controller **200** may include a current calculator **220A** calculating currents $CUR(X-1)$ of the previous frame and currents $CUR(X)$ of the present frame for the display blocks for the each colors, a current storer **240A** storing the currents $CUR(X-1)$ of the previous frame for the display blocks for the each colors and a static image determiner **260** receiving the currents $CUR(X)$ of the present frame for the display blocks for the each colors from the current calculator **220A** and the currents $CUR(X-1)$ of the previous frame for the display blocks for the each colors from the current storer **240A** and determining the image displayed on the display panel as a static image based on the differences between the currents $CUR(X-1)$ of the previous frame for the display blocks for the each colors and the currents $CUR(X)$ of the present frame for the display blocks for the each colors.

[0122] The driving controller **200** may further include an image compensator **280** receiving a flag signal FL indicating the image displayed on the display pane is the static image from the static image determiner **260** and reduce the luminance for the input image data $IMG(X)$.

[0123] For example, the driving controller **200** may convert the grayscale value of the input image data IMG to the current by a gamma conversion. The current of the input image data IMG corresponds to a luminance actually perceived by a user's eyes. Accordingly, when the image displayed on the display panel is determined as the static image using the current of the input image data IMG , the accuracy of the determination of the static image may be enhanced compared to a case in which the static image is determined using the grayscale value of the input image data IMG .

[0124] According to the present embodiment, the image displayed on the display panel is determined as the static image based on the differences between the currents $CUR(X-1)$ of the previous frame and the currents $CUR(X)$ of the present frame for the display blocks for the each colors to enhance an accuracy of the determination of the static image.

[0125] Accordingly, an error mistakenly determining a moving image with little movements between frames and a moving image which is dark overall as a static image may be prevented.

[0126] Therefore, when the same image is displayed on the display panel for a long time, the luminance of the display image may reduce so that the afterimage of the display panel **100** may be prevented and the power consumption of the display apparatus may be reduced.

[0127] FIG. **16** is a block diagram illustrating a driving controller **200** of a display apparatus according to an embodiment of the present inventive concept.

[0128] The display apparatus according to the present embodiment is substantially the same as the display apparatus of the previous embodiment explained referring to FIGS. **1** to **14** except that the driving controller determines the image displayed on the display panel as the static image not based on the loads of the frames but based on luminances of the frames. Thus, the same reference numerals will be used to refer to the same or like parts as those described in the previous embodiment of FIGS. **1** to **14** and any repetitive explanation concerning the above elements will be omitted.

[0129] Referring to FIGS. **1** to **8**, **10** to **14** and **16**, the driving controller **200** may include a luminance calculator **220B** calculating luminances $LUM(X-1)$ of the previous frame and luminances $LUM(X)$ of the present frame for the display blocks for the each colors, a luminance storer **240B** storing the luminances $LUM(X-1)$ of the previous frame for the display blocks for the each colors and a static image determiner **260** receiving the luminances $LUM(X)$ of the present frame for the display blocks for the each colors from the luminance calculator **220B** and the luminances $LUM(X-1)$ of the previous frame for the display blocks for the each colors from the luminance storer **240B** and determining the image displayed on the display panel as a static image based on the differences between the luminances $LUM(X-1)$ of the previous frame for the display

blocks for the each colors and the luminances LUM(X) of the present frame for the display blocks for the each colors.

[0130] The driving controller **200** may further include an image compensator **280** receiving a flag signal FL indicating the image displayed on the display panel as the static image from the static image determiner **260** and operating the luminance decrease for the input image data IMG(X).

[0131] For example, the driving controller **200** may convert the grayscale value of the input image data IMG to the luminance by a gamma conversion. The luminance of the input image data IMG is actually perceived by a user's eyes. Accordingly, when the image displayed on the display panel is determined as the static image using the luminance of the input image data IMG, the accuracy of the determination of the static image may be enhanced compared to a case in which the static image is determined using the grayscale value of the input image data IMG.

[0132] According to the present embodiment, the static image may be determined based on the differences between the luminances LUM(X-1) of the previous frame and the luminances LUM(X) of the present frame for the display blocks of the display panel **100** for the each colors to enhance an accuracy of the determination of the static image.

[0133] Accordingly, an error mistakenly determining a moving image with little movements between frames and a moving image which is dark overall as a static image may be prevented.

[0134] Therefore, when the same image is displayed on the display panel for a long time, the luminance of the display image may reduce so that the afterimage of the display panel **100** may be prevented and the power consumption of the display apparatus may be reduced.

[0135] FIG. **17** is a block diagram illustrating an electronic apparatus **1000** according to an embodiment of the present inventive concept. FIG. **18** is a diagram illustrating an example in which the electronic apparatus **1000** of FIG. **17** is implemented as a smartphone. FIG. **19** is a diagram illustrating an example in which the electronic apparatus **1000** of FIG. **17** is implemented as a monitor.

[0136] Referring to FIGS. **17** to **19**, the electronic apparatus **1000** may include a processor **1010**, a memory device **1020**, a storage device **1030**, an input/output (I/O) device **1040**, a power supply **1050**, and a display apparatus **1060**. Here, the display apparatus **1060** may be the display apparatus of FIG. **1**. In addition, the electronic apparatus **1000** may further include a plurality of ports for communicating with a video card, a sound card, a memory card, a universal serial bus (USB) device, other electronic apparatuses, etc.

[0137] In an embodiment, as illustrated in FIG. **18**, the electronic apparatus **1000** may be implemented as a smartphone. In an embodiment, as illustrated in FIG. **19**, the electronic apparatus **1000** may be implemented as a monitor. However, the electronic apparatus **1000** is not limited thereto. For example, the electronic apparatus **1000** may be implemented as a television, a cellular phone, a video phone, a smart pad, a smart watch, a tablet PC, a car navigation system, a laptop, a head mounted display (HMD) device, and the like.

[0138] The processor **1010** may perform various computing functions or various tasks. The processor **1010** may be a micro-processor, a central processing unit (CPU), an application processor (AP), and the like. The processor **1010** may be coupled to other components via an address bus, a control bus, a data bus, etc. Further, the processor **1010** may be coupled to an extended bus such as a peripheral component interconnection (PCI) bus.

[0139] The processor **1010** may output the input image data IMG and the input control signal CONT to the driving controller **200** of FIG. **1**.

[0140] The memory device **1020** may store data for operations of the electronic apparatus **1000**. For example, the memory device **1020** may include at least one non-volatile memory device such as an erasable programmable read-only memory (EPROM) device, an electrically erasable programmable read-only memory (EEPROM) device, a flash memory device, a phase change random access memory (PRAM) device, a resistance random access memory (RRAM) device, a nano floating gate memory (NFGM) device, a polymer random access memory (PoRAM) device, a

magnetic random access memory (MRAM) device, a ferroelectric random access memory (FRAM) device, and the like and/or at least one volatile memory device such as a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, a mobile DRAM device, and the like.

[0141] The storage device **1030** may include a solid state drive (SSD) device, a hard disk drive (HDD) device, a CD-ROM device, and the like. The I/O device **1040** may include an input device such as a keyboard, a keypad, a mouse device, a touch-pad, a touch-screen, and the like and an output device such as a printer, a speaker, and the like. In some embodiments, the display apparatus **1060** may be included in the I/O device **1040**. The power supply **1050** may provide power for operations of the electronic apparatus **1000**. The display apparatus **1060** may be coupled to other components via the buses or other communication links.

[0142] FIG. **20** is a block diagram illustrating an electronic apparatus **101** according to an embodiment of the present inventive concept.

[0143] Referring to FIGS. **1** to **20**, an electronic apparatus **101** outputs various information through a display module **140** in an operating system. When a processor **110** executes an application stored in a memory **120**, the display module **140** provides application information to a user through a display panel **141**.

[0144] The processor **110** obtains an external input through an input module **130** or a sensor module **161** and executes an application corresponding to the external input. For example, when the user selects a camera icon displayed on the display panel **141**, the processor **110** obtains a user input through an input sensor **161-2** and activates a camera module **171**. The processor **110** transfers image data corresponding to a captured image obtained through the camera module **171** to the display module **140**. The display module **140** may display an image corresponding to the captured image through the display panel **141**.

[0145] In an embodiment, when a personal information authentication is executed in the display module **140**, a fingerprint sensor **161-1** obtains input fingerprint information as input data. The processor **110** compares input data obtained through the fingerprint sensor **161-1** with authentication data stored in the memory **120**, and executes an application according to a comparison result. The display module **140** may display information executed according to application logic through the display panel **141**.

[0146] In an embodiment, when a music streaming icon displayed on the display module **140** is selected, the processor **110** obtains a user input through the input sensor **161-2** and activates a music streaming application stored in the memory **120**. When a music execution command is input in the music streaming application, the processor **110** activates a sound output module **163** to provide sound information corresponding to the music execution command to the user.

[0147] In the above, the operation of the electronic apparatus **101** is briefly described. Hereinafter, a configuration of the electronic apparatus **101** is described in detail. Some of elements of the electronic apparatus **101** described later may be integrated and provided as one element, or one element may be separated as two or more elements.

[0148] The electronic apparatus **101** may communicate with an external electronic apparatus **102** through a network (e.g., a short-range wireless communication network or a long-range wireless communication network). According to an embodiment, the electronic apparatus **101** may include the processor **110**, the memory **120**, the input module **130**, the display module **140**, a power module **150**, an embedded module **160**, and an external module **170**. According to an embodiment, in the electronic apparatus **101**, at least one of the above-described elements may be omitted or one or more other apparatus may be added. According to an embodiment, some of the above-described elements (e.g., the sensor module **161**, an antenna module **162** or the sound output module **163**) may be integrated into another element (e.g., the display module **140**).

[0149] The processor **110** may execute software to control at least one other element (e.g., hardware or software element) of the electronic apparatus **101** connected to the processor **110** and

to perform various data processing or operations. According to an embodiment, as at least part of the data processing or the operations, the processor **110** may store receive instructions or data from other elements (e.g. the input module **130**, the sensor module **161** or a communication module **173**) in a volatile memory **121**, may process the instructions or data stored in the volatile memory **121** and may store result data of the processing in a nonvolatile memory **122**.

[0150] The processor **110** may include a main processor **111** and an auxiliary processor **112**. The main processor **111** may include at least one of a central processing unit (CPU) **111-1** and an application processor (AP). The main processor **111** may further include any one or more of a graphic processing unit (GPU) **111-2**, a communication processor (CP) and an image signal processor (ISP). The main processor **111** may further include a neural processing unit (NPU) **111-3**. The neural network processing unit **111-3** is a processor specialized in processing an artificial intelligence model. The artificial intelligence model may be generated through a machine learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be one of a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN) and a deep Q-networks or a combination of two or more of the above. However, the artificial neural network is not limited to the above examples. The artificial intelligence model may include software structures, in addition to hardware structures or instead of the hardware structures. At least two of the above-described processing units and the above-described processors may be implemented as an integrated element (e.g., a single chip) or each may be implemented as independent elements (e.g., in a plurality of chips).

[0151] The auxiliary processor **112** may include a controller. The controller may include an interface conversion circuit and a timing control circuit. The controller receives an image signal from the main processor **111**, converts a data format of the image signal to meet interface specifications with the display module **140**, and outputs image data. The controller may output various control signals for driving the display module **140**.

[0152] The auxiliary processor **112** may further include a data converting circuit **112-2**, a gamma correction circuit **112-3** and a rendering circuit **112-4**. The data converting circuit **112-2** may receive the image data from the controller and may compensate the image data such that the image is displayed with a desired luminance according to characteristics of the electronic apparatus **101** or a user setting or may convert the image data to reduce a power consumption or compensate for afterimages. The gamma correction circuit **112-3** may convert the image data or a gamma reference voltage such that the image displayed on the electronic apparatus **101** has desired gamma characteristics. The rendering circuit **112-4** may receive the image data from the controller and may render the image data based on a pixel arrangement of the display panel **141** included in the electronic apparatus **101**. At least one of the data converting circuit **112-2**, the gamma correction circuit **112-3** and the rendering circuit **112-4** may be integrated into another element (e.g. the main processor **111** or the controller). At least one of the data converting circuit **112-2**, the gamma correction circuit **112-3** and the rendering circuit **112-4** may be integrated into a data driver **143** to be described later.

[0153] The memory **120** may store various data used by at least one element (e.g., the processor **110** or the sensor module **161**) of the electronic apparatus **101** and input data or output data for commands related thereto. The memory **120** may include at least one of the volatile memory **121** and the nonvolatile memory **122**.

[0154] The input module **130** may receive commands or data used to the elements (e.g., the processor **110**, the sensor module **161** or the sound output module **163**) of the electronic apparatus **101** from the outside of the electronic apparatus **101** (e.g., the user or the external electronic apparatus **102**).

[0155] The input module **130** may include a first input module **131** for receiving commands or data

from the user and a second input module **132** for receiving commands or data from the external electronic apparatus **102**. The first input module **131** may include a microphone, a mouse, a keyboard, a key (e.g., a button) or a pen (e.g., a passive pen or an active pen). The second input module **132** may support a designated protocol capable of connecting to the external electronic apparatus **102** by wire or wirelessly. According to an embodiment, the second input module **132** may include a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface or an audio interface. The second input module **132** may include a connector physically connected to the external electronic apparatus **102**, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0156] The display module **140** visually provides information to the user. The display module **140** may include the display panel **141**, a scan driver **142** and the data driver **143**. The display module **140** may further include a window, a chassis and a bracket to protect the display panel **141**.

[0157] The display panel **141** may include a liquid crystal display panel, an organic light emitting display panel or an inorganic light emitting display panel. A type of the display panel **141** is not particularly limited. The display panel **141** may be a rigid type or a flexible type capable of being rolled or folded. The display module **140** may further include a supporter or a heat dissipation member supporting the display panel **141**.

[0158] The scan driver **142** may be mounted on the display panel **141** as a driving chip.

Alternatively, the scan driver **142** may be integrated on the display panel **141**. For example, the scan driver **142** may include an amorphous silicon TFT gate driver circuit (ASG) integrated on the display panel **141**, a low temperature polycrystalline silicon (LTPS) TFT gate driver circuit integrated on the display panel **141**, or an oxide semiconductor TFT gate driver circuit (OSG) integrated on the display panel **141**. The scan driver **142** receives a control signal from the controller and outputs the scan signals to the display panel **141** in response to the control signal.

[0159] The display module **140** may further include a light emission driver. The light emission driver outputs a light emission control signal to the display panel **141** in response to a control signal received from the controller. The light emission driver may be formed independently from the scan driver **142**. Alternatively, the light emission driver and the scan driver **142** may be integrally formed.

[0160] The data driver **143** receives a control signal from the controller and converts the image data into an analog voltage (e.g., the data voltage) and output the data voltages to the display panel **141** in response to the control signal.

[0161] The data driver **143** may be integrated into another element (e.g., the controller). The functions of the interface conversion circuit and the timing control circuit of the controller described above may be integrated into the data driver **143**.

[0162] The display module **140** may further include a voltage generating circuit. The voltage generating circuit may output various voltages for driving the display panel **141**.

[0163] The power module **150** supplies power to elements of the electronic apparatus **101**. The power module **150** may include a battery which supplies a power voltage. The battery may include a non-rechargeable primary cell, a rechargeable secondary cell or a fuel cell. The power module **150** may include a power management integrated circuit (PMIC). The PMIC supplies optimized power to each of the above-described modules and modules described later. The power module **150** may include a wireless power transmission/reception member electrically connected to the battery. The wireless power transmission/reception member may include a plurality of antenna radiators in a form of coils.

[0164] The electronic apparatus **101** may further include the embedded module **160** and the external module **170**. The embedded module **160** may include the sensor module **161**, the antenna module **162** and the sound output module **163**. The external module **170** may include the camera module **171**, a light module **172** and the communication module **173**.

[0165] The sensor module **161** may detect an input by a user's body or an input by the pen among

the first input module **131**, and generate an electrical signal or data value corresponding to the input. The sensor module **161** may include at least one of the fingerprint sensor **161-1**, the input sensor **161-2** and a digitizer **161-3**.

[0166] The fingerprint sensor **161-1** may generate a data value corresponding to a user's fingerprint. The fingerprint sensor **161-1** may include one of an optical fingerprint sensor or a capacitive fingerprint sensor.

[0167] The input sensor **161-2** may generate data values corresponding to coordinate information of the input by the user's body or the input by the pen. The input sensor **161-2** generates a capacitance change due to an input as a data value. The input sensor **161-2** may detect an input by the passive pen or transmit/receive data to/from the active pen.

[0168] The input sensor **161-2** may measure biosignals such as a blood pressure, a moisture, or a body fat. For example, when a user touches a part of his body to a sensor layer or a sensing panel and does not move for a certain period of time, the input sensor **161-2** may detect the biosignal based on a change in an electric field caused by the part of the body so that the display module **140** may output user's desired information.

[0169] The digitizer **161-3** may generate a data value corresponding to the coordinate information input by the pen. The digitizer **161-3** generates an amount of electromagnetic change by the input as a data value. The digitizer **161-3** may detect an input by the passive pen or transmit/receive data to/from the active pen.

[0170] At least one of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** may be formed as a sensor layer on the display panel **141** through a continuous process. The fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** may be disposed on the display panel **141**. At least one of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3**, for example, the digitizer **161-3**, may be disposed under the display panel **141**.

[0171] At least two or more of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** may be integrated into the sensing panel through the same process. When at least two or more of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** are integrated into the sensing panel, the sensing panel may be disposed between the display panel **141** and a window disposed over an upper surface of the display panel **141**. According to an embodiment, the sensing panel may be disposed on the window. The present inventive concept may not be limited to a position of the sensing panel.

[0172] At least one of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** may be embedded in the display panel **141**. For example, at least one of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** is formed simultaneously with the display panel **141** through a process of forming elements included in the display panel **141** (e.g., light emitting elements, transistors, etc.).

[0173] In addition, the sensor module **161** may generate an electrical signal or a data value corresponding to an internal state or an external state of the electronic apparatus **101**. For example, the sensor module **161** may further include a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an IR (infrared) sensor, a biosensor, a temperature sensor, a humidity sensor or an illuminance sensor.

[0174] The antenna module **162** may include one or more antennas for transmitting a signal or power to outside or receiving a signal or power from outside. According to an embodiment, the communication module **173** may transmit a signal to an external electronic apparatus or receive a signal from an external electronic apparatus through an antenna suitable for a communication method. An antenna pattern of the antenna module **162** may be integrated with an element of the display module **140** (e.g., the display panel **141**) or the input sensor **161-2**.

[0175] The sound output module **163** is a device for outputting sound signals to the outside of the electronic apparatus **101**. For example, the sound output module **163** may include a speaker used

for general purposes such as playing multimedia or recording and a receiver used exclusively for receiving a call. According to an embodiment, the receiver may be formed integrally with or separately from the speaker. A sound output pattern of the sound output module **163** may be integrated with the display module **140**.

[0176] The camera module **171** may capture still images and moving images. According to an embodiment, the camera module **171** may include one or more lenses, an image sensor or an image signal processor. The camera module **171** may further include an infrared camera capable of determining a presence or an absence of a user, the user's location and the user's gaze.

[0177] The light module **172** may provide a light. The light module **172** may include a light emitting diode or a xenon lamp. The light module **172** may operate in conjunction with the camera module **171** or operate independently.

[0178] The communication module **173** may support establishment of a wired or wireless communication channel between the electronic apparatus **101** and the external electronic apparatus **102** and communication through the established communication channel. The communication module **173** may include one or both of a wireless communication module such as a cellular communication module, a short-distance wireless communication module, or a global navigation satellite system (GNSS) communication module and a wired communication module such as a local area network (LAN) communication module, or a power line communication module. The communication module **173** may communicate with the external electronic apparatus **102** through a short-range communication network such as Bluetooth, WiFi direct or infrared data association (IrDA) or a long-distance communication network such as a cellular network, the Internet, or a computer network (e.g., LAN or WAN). The various types of communication modules **173** described above may be implemented as a single chip or may be implemented as separate chips.

[0179] The input module **130**, the sensor module **161** and the camera module **171** may be used to control the operation of the display module **140** in conjunction with the processor **110**.

[0180] The processor **110** outputs commands or data to the display module **140**, the sound output module **163**, the camera module **171** or the light module **172** based on the input data received from the input module **130**. For example, the processor **110** may generate image data corresponding to input data applied through a mouse or an active pen, and output the generated image data to the display module **140** or the processor **110** may generate command data corresponding to the input data and output the generated command data to the camera module **171** or the light module **172**.

When input data is not received from the input module **130** for a certain period of time, the processor **110** converts an operation mode of the electronic apparatus **101** into a low power mode or a sleep mode so that a power consumption of the electronic apparatus **101** may be reduced.

[0181] The processor **110** outputs commands or data to the display module **140**, the sound output module **163**, the camera module **171** or the light module **172** based on sensed data received from the sensor module **161**. For example, the processor **110** may compare authentication data applied by the fingerprint sensor **161-1** with authentication data stored in the memory **120**, and then execute an application according to the comparison result. The processor **110** may execute commands or output corresponding image data to the display module **140** based on the sensed data sensed by the input sensor **161-2** or the digitizer **161-3**. When the sensor module **161** includes a temperature sensor, the processor **110** may receive temperature data for the temperature measured from the sensor module **161** and may further perform luminance correction on the image data based on the temperature data.

[0182] The processor **110** may receive determined data about the presence or the absence of the user, the user's location and the user's gaze from the camera module **171**. The processor **110** may further perform luminance correction on the image data based on the determined data. For example, the processor **110**, which determines the presence or the absence of the user through an input from the camera module **171**, may display image data having the luminance corrected by the data converting circuit **112-2** or the gamma correction circuit **112-3** to the display module **140**.

[0183] Some of the above elements may be connected to each other through a communication method between peripheral devices such as a bus, a general purpose input/output (GPIO), a serial peripheral interface (SPI), a mobile industry processor interface (MIPI), or an ultra-path interconnect (UPI) link to exchange signals (e.g., commands or data) with each other. The processor **110** may communicate with the display module **140** through an agreed interface. For example, the processor **110** may communicate with the display module **140** through any one of the above communication methods. The present invention may not be limited to the above communication methods.

[0184] The electronic apparatus **101** according to various embodiments disclosed in the disclosure may be various types of apparatuses. For example, the electronic apparatus **101** may include at least one of a monitor, a portable communication apparatus (e.g., a smart phone), a computer apparatus, a portable multimedia apparatus, a portable medical apparatus, a camera, a wearable device and a home appliance. The electronic apparatus **101** according to the embodiment of the disclosure may not be limited to the aforementioned apparatuses.

[0185] For example, the display panel **100** of FIG. **1** may correspond to the display panel **141** of FIG. **20**. For example, the driving controller **200** of FIG. **1** may correspond to the controller of the auxiliary processor **112** of FIG. **20**. For example, the gate driver **300** of FIG. **1** may correspond to the scan driver **142** of FIG. **20**. For example, the data driver **500** of FIG. **1** may correspond to the data driver **143** of FIG. **20**.

[0186] According to the embodiments of the display apparatus, the method of driving the display panel using the display apparatus and the electronic apparatus including the display apparatus, the accuracy of the determination of the static image may be enhanced.

[0187] The foregoing is illustrative of the present inventive concept and is not to be construed as limiting thereof. Although a few embodiments of the present inventive concept have been described, those skilled in the art will readily appreciate that many modifications are possible in the embodiments without materially departing from the novel teachings and advantages of the present inventive concept. Accordingly, all such modifications are intended to be included within the scope of the present inventive concept as defined in the claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents but also equivalent structures. Therefore, it is to be understood that the foregoing is illustrative of the present inventive concept and is not to be construed as limited to the specific embodiments disclosed, and that modifications to the disclosed embodiments, as well as other embodiments, are intended to be included within the scope of the appended claims. The present inventive concept is defined by the following claims, with equivalents of the claims to be included therein.

Claims

1. A display apparatus comprising: a display panel including a plurality of display blocks; a data driver configured to output data voltages to the display panel; and a driving controller configured to control the data driver, wherein the driving controller is configured to calculate differences between loads of a previous frame and loads of a present frame for the plurality of display blocks for each colors, to determine whether an image is a static image or not based on the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors, and to reduce luminance of the image when the image is determined as the static image.
2. The display apparatus of claim 1, wherein the driving controller is configured to determine a maximum value among absolute values of the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors.
3. The display apparatus of claim 2, wherein, when a load of an X-1-th frame of a first color in a

first display block is $\text{LOAD}(X-1, R1)$, a load of an X-th frame of the first color in the first display block is $\text{LOAD}(X, R1)$, a load of the X-1-th frame of the first color in a second display block is $\text{LOAD}(X-1, R2)$, a load of the X-th frame of the first color in the second display block is $\text{LOAD}(X, R2)$, a load of the X-1-th frame of the first color in an N-th display block is $\text{LOAD}(X-1, RN)$, a load of the X-th frame of the first color in the N-th display block is $\text{LOAD}(X, RN)$, a load of the X-1-th frame of a second color in the first display block is $\text{LOAD}(X-1, G1)$, a load of the X-th frame of the second color in the first display block is $\text{LOAD}(X, G1)$, a load of the X-1-th frame of the second color in the second display block is $\text{LOAD}(X-1, G2)$, a load of the X-th frame of the second color in the second display block is $\text{LOAD}(X, G2)$, a load of the X-1-th frame of the second color in the N-th display block is $\text{LOAD}(X-1, GN)$, a load of the X-th frame of the second color in the N-th display block is $\text{LOAD}(X, GN)$, a load of the X-1-th frame of a third color in the first display block is $\text{LOAD}(X-1, B1)$, a load of the X-th frame of the third color in the first display block is $\text{LOAD}(X, B1)$, a load of the X-1-th frame of the third color in the second display block is $\text{LOAD}(X-1, B2)$, a load of the X-th frame of the third color in the second display block is $\text{LOAD}(X, B2)$, a load of the X-1-th frame of the third color in the N-th display block is $\text{LOAD}(X-1, BN)$, a load of the X-th frame of the third color in the N-th display block is $\text{LOAD}(X, BN)$, the maximum value is $\text{MAX}(|\text{LOAD}(X-1, R1)-\text{LOAD}(X, R1)|, |\text{LOAD}(X-1, R2)-\text{LOAD}(X, R2)|, \dots, |\text{LOAD}(X-1, RN)-\text{LOAD}(X, RN)|, |\text{LOAD}(X-1, G1)-\text{LOAD}(X, G1)|, |\text{LOAD}(X-1, G2)-\text{LOAD}(X, G2)|, \dots, |\text{LOAD}(X-1, GN)-\text{LOAD}(X, GN)|, |\text{LOAD}(X-1, B1)-\text{LOAD}(X, B1)|, |\text{LOAD}(X-1, B2)-\text{LOAD}(X, B2)|, \dots, |\text{LOAD}(X-1, BN)-\text{LOAD}(X, BN)|)$.

4. The display apparatus of claim 2, wherein the driving controller is configured to generate a sensitivity value by normalizing the maximum value to a ratio of a difference between a maximum load and a minimum load.

5. The display apparatus of claim 4, wherein, when the sensitivity value is Δ , an absolute value of a difference between a load of an X-1-th frame of a first color in a first display block and a load of an X-th frame of the first color in the first display block is $D(R1)$, an absolute value of a difference between a load of the X-1-th frame of the first color in a second display block and a load of the X-th frame of the first color in the second display block is $D(R2)$, an absolute value of a difference between a load of the X-1-th frame of the first color in an N-th display block and a load of the X-th frame of the first color in the N-th display block is $D(RN)$, an absolute value of a difference between a load of the X-1-th frame of a second color in the first display block and a load of the X-th frame of the second color in the first display block is $D(G1)$, an absolute value of a difference between a load of the X-1-th frame of the second color in the second display block and a load of the X-th frame of the second color in the second display block is $D(G2)$, an absolute value of a difference between a load of the X-1-th frame of the second color in the N-th display block and a load of the X-th frame of the second color in the N-th display block is $D(GN)$, an absolute value of a difference between a load of the X-1-th frame of a third color in the first display block and a load of the X-th frame of the third color in the first display block is $D(B1)$, an absolute value of a difference between a load of the X-1-th frame of the third color in the second display block and a load of the X-th frame of the third color in the second display block is $D(B2)$, an absolute value of a difference between a load of the X-1-th frame of the third color in the N-th display block and a load of the X-th frame of the third color in the N-th display block is $D(BN)$, the maximum load is MAXLO , the minimum load is MINLO and a number of the plurality of display blocks of the

display panel is N,
$$= \frac{\text{MAX}(D(R1), D(R2), \dots, D(RN), D(G1), D(G2), \dots, D(GN), D(B1), D(B2), \dots, D(BN))}{(\text{MAXLO} - \text{MINLO}) / (N \times 3)}$$
 is satisfied.

6. The display apparatus of claim 4, wherein, when the sensitivity value is less than a first threshold value, the driving controller is configured to determine data of the present frame as static image data.

7. The display apparatus of claim 6, wherein, when a maintaining time of the static image is greater

than a second threshold value, the driving controller is configured to reduce the luminance of the image.

8. The display apparatus of claim 1, wherein the driving controller comprises: a load calculator configured to calculate the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors; a load storer configured to store the loads of the previous frame for the plurality of display blocks for the each colors; and a static image determiner configured to receive the loads of the present frame for the plurality of display blocks for the each colors from the load calculator and the loads of the previous frame for the plurality of display blocks for the each colors from the load storer and to determine whether the image is the static image or not based on the differences between the loads of the previous frame for the plurality of display blocks for the each colors and the loads of the present frame for the plurality of display blocks for the each colors.

9. The display apparatus of claim 8, wherein the driving controller further comprises an image compensator configured to receive a flag signal indicating the image as the static image from the static image determiner and to reduce the luminance when the image is determined as the static image.

10. A method of driving a display panel, the method comprising calculating differences between loads of a previous frame and loads of a present frame for display blocks for each colors in the display panel; determining whether an image is a static image or not based on the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors; reducing luminance of the image when the image is determined as the static image to generate compensated image data; and generating data voltages based on the compensated image data and outputting the data voltages to the display panel.

11. The method of claim 10, wherein the determining whether the image is a static image or not comprises: determining a maximum value among absolute values of the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors.

12. The method of claim 11, wherein the determining whether the image is a static image or not further comprises: generating a sensitivity value by normalizing the maximum value to a ratio of a difference between a maximum load and a minimum load.

13. The method of claim 12, wherein the determining whether the image is a static image or not further comprises: determining data of the present frame as static image data when the sensitivity value is less than a first threshold value.

14. The method of claim 13, wherein the reducing luminance of the image when the imaged is determined as the static image comprises: reducing the luminance of the image when a maintaining time of the static image is greater than a second threshold value.

15. A display apparatus comprising: a display panel including a plurality of display blocks; a data driver configured to output data voltages to the display panel; and a driving controller configured to control the data driver, wherein the driving controller is configured to calculate differences between currents or luminances of a previous frame and currents or luminances of a present frame for the plurality of display blocks for each colors, to determine whether an image is a static image or not based on the differences between the currents or luminances of the previous frame and the currents or luminances of the present frame for the plurality of display blocks for the each colors, and to reduce luminance of the image when the image is determined as the static image.

16. The display apparatus of claim 15, wherein the driving controller is configured to determine a maximum value among absolute values of the differences between the currents or luminances of the previous frame and the currents or luminances of the present frame for the plurality of display blocks for the each colors.

17. The display apparatus of claim 16, wherein the driving controller is configured to generate a sensitivity value by normalizing the maximum value to a ratio of a difference between a maximum

current or luminance and a minimum current or luminance.

18. The display apparatus of claim 17, wherein, when the sensitivity value is less than a first threshold value, the driving controller is configured to determine data of the present frame as static image data.

19. The display apparatus of claim 15, wherein the driving controller comprises: a current calculator configured to calculate the currents of the previous frame and the currents of the present frame for the plurality of display blocks for the each colors; a current storer configured to store the currents of the previous frame for the plurality of display blocks for the each colors; and a static image determiner configured to receive the currents of the present frame for the plurality of display blocks for the each colors from the current calculator and the currents of the previous frame for the plurality of display blocks for the each colors from the current storer and to determine the image as the static image based on the differences between the currents of the previous frame for the plurality of display blocks for the each colors and the currents of the present frame for the plurality of display blocks for the each colors.

20. The display apparatus of claim 15, wherein the driving controller comprises: a luminance calculator configured to calculate the luminances of the previous frame and the luminances of the present frame for the plurality of display blocks for the each colors; a luminance storer configured to store the luminances of the previous frame for the plurality of display blocks for the each colors; and a static image determiner configured to receive the luminances of the present frame for the plurality of display blocks for the each colors from the luminance calculator and the luminances of the previous frame for the plurality of display blocks for the each colors from the luminance storer and to determine the image as the static image based on the differences between the luminances of the previous frame for the plurality of display blocks for the each colors and the luminances of the present frame for the plurality of display blocks for the each colors.

21. An electronic apparatus comprising: a display panel including a plurality of display blocks; a data driver configured to output data voltages to the display panel; a driving controller configured to control the data driver; and a processor configured to output input image data and an input control signal to the driving controller, wherein the driving controller is configured to calculate differences between loads of a previous frame and loads of a present frame for the plurality of display blocks for each colors, to determine whether an image is a static image or not based on the differences between the loads of the previous frame and the loads of the present frame for the plurality of display blocks for the each colors, and to reduce luminance of the image when the image is determined as the static image.
